

Studio di funzione - Esercizio

giovedì 12 dicembre 2024 18:02

$$y = x^3 - x^2$$

1. DOMINIO $D = \mathbb{R} \quad D = (-\infty, +\infty)$

2. SIMMETRIE

$$f(-x) \stackrel{?}{=} f(x) \rightarrow \begin{array}{l} (-x)^3 - (-x)^2 \\ -x^3 - x^2 \end{array} \stackrel{?}{=} \begin{array}{l} x^3 - x^2 \\ x^3 - x^2 \end{array} \quad \text{NON È PARI}$$

$$f(-x) \stackrel{?}{=} -f(x) \rightarrow \begin{array}{l} -x^3 - x^2 \\ -x^3 - x^2 \end{array} \stackrel{?}{=} \begin{array}{l} -(x^3 - x^2) \\ -x^3 + x^2 \end{array} \quad \text{NON È DISPARE}$$

3. INTERSEZIONE

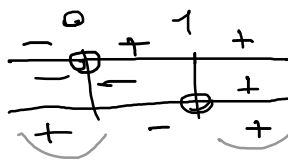
$$\begin{cases} x^3 - x^2 = 0 \\ y = 0 \end{cases} \rightarrow \begin{cases} \overset{F_1}{x^2}(\overset{F_2}{x-1}) = 0 \\ y = 0 \end{cases} \rightarrow \begin{cases} \begin{array}{l} \nearrow F_1 \quad x^2 = 0 \\ \searrow F_2 \quad x-1 = 0 \end{array} \\ y = 0 \end{cases}$$

$$\begin{cases} y = x^3 - x^2 \\ x = 0 \end{cases} \rightarrow \begin{cases} y = 0 \\ x = 0 \end{cases} \quad \boxed{(0,0)}$$

4. SEGNO

$$x^3 - x^2 > 0 \rightarrow \begin{array}{c} \overset{F_1}{x^2}(\overset{F_2}{x-1}) > 0 \end{array}$$

$$\begin{array}{l} F_1 \quad x^2 > 0 \rightarrow x > 0 \\ F_2 \quad x-1 > 0 \rightarrow x > 1 \end{array}$$



$$S = (-\infty, 0) \cup (1, +\infty)$$

$$I_p = S ; I$$

5. ASINTOTI

$$D = ([-\infty], [+ \infty])$$

$$\lim_{x \rightarrow -\infty} x^3 - x^2 = -\infty - (-\infty)^2 = -\infty - \infty = -\infty$$

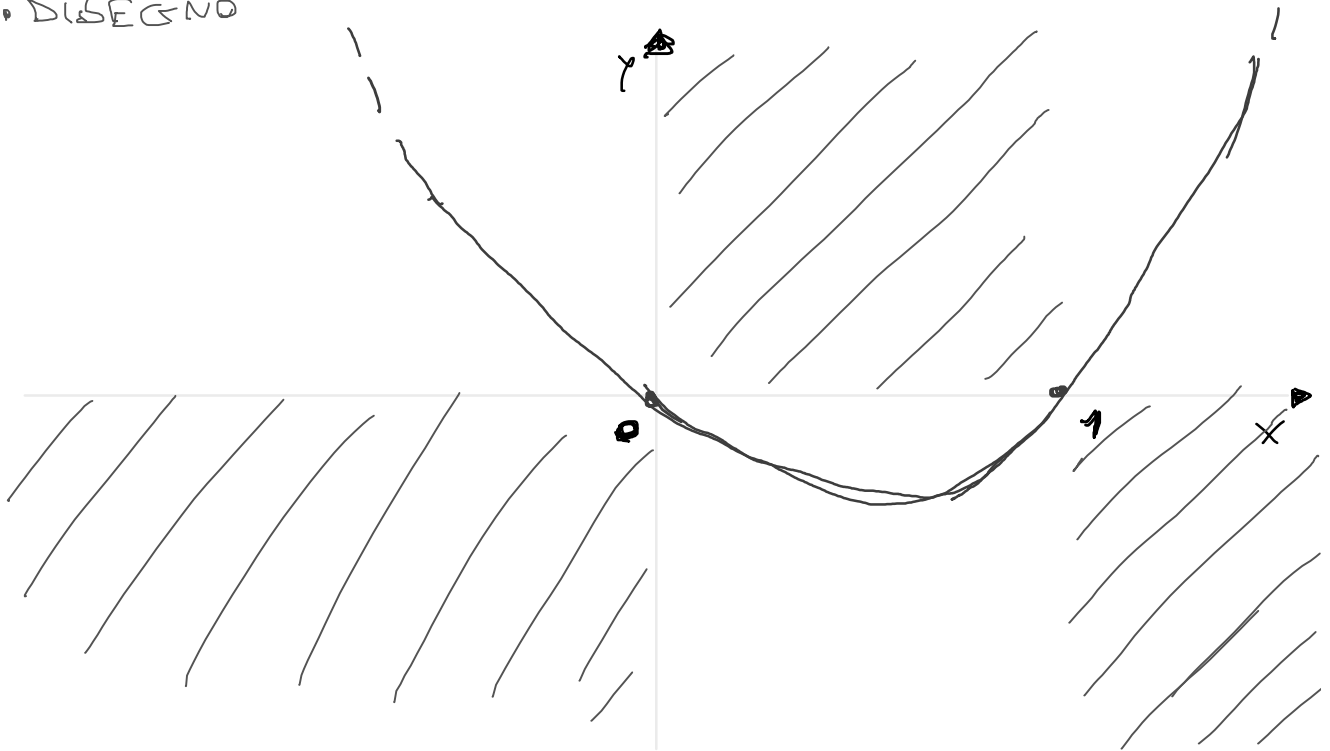
$$\lim_{x \rightarrow +\infty} x^3 - x^2 = +\infty - \infty \quad \text{p.i.}$$

$$x \rightarrow +\infty$$

$$\lim_{x \rightarrow +\infty} x^3 \left(1 - \frac{1}{x}\right) = +\infty (1) = +\infty \quad \text{NO ASI}$$

$$m = \lim_{x \rightarrow \pm\infty} \frac{x^3 - x^2}{x} = \lim_{x \rightarrow \pm\infty} \frac{x^2 (1 - \frac{1}{x})}{x} = +\infty$$

6. DISEGNO



$$y = -x^3 + x^2$$

1. DOMINIO $D = \mathbb{R} \quad D = (-\infty, +\infty)$

2. SIMMETRIA

$$f(-x) = f(x) \rightarrow \begin{aligned} -(-x)^3 + (-x)^2 & \stackrel{?}{=} -x^3 + x^2 \\ +x^3 + x^2 & = -x^3 + x^2 \quad \text{NO.} \end{aligned}$$

$$f(-x) = -f(x) \rightarrow \begin{aligned} x^3 + x^2 & = -(-x^3 + x^2) \\ x^3 + x^2 & = +x^3 - x^2 \quad \text{NO.} \end{aligned}$$

3. INTERSEZIONI

$$(-x^3 + x^2 = 0 \quad \vee \quad x^3 - x^2 = 0 \quad \vee \quad x^2 (x - 1) = 0$$

$$\begin{cases} Y=0 \end{cases} \rightarrow \begin{cases} Y=0 \end{cases} \rightarrow \begin{cases} Y=0 \end{cases}$$

$$\begin{aligned} F_1 \quad x^2 &= 0 \rightarrow x=0 \\ F_2 \quad x-1 &= 0 \rightarrow x=1 \end{aligned}$$

$$\begin{pmatrix} 0, 0 \\ 1, 0 \end{pmatrix}$$

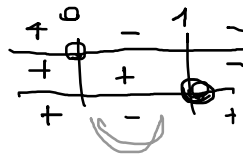
$$\begin{cases} -x^3 + x^2 = Y \\ x = 0 \end{cases} \rightarrow \begin{cases} Y=0 \\ x=0 \end{cases} \quad (0, 0)$$

4. SEGNO

$$-x^3 + x^2 > 0 \rightarrow x^3 - x^2 < 0 \rightarrow x^2 (x-1) < 0$$

$$F_1 \quad x^2 < 0 \rightarrow x < 0$$

$$F_2 \quad x-1 < 0 \rightarrow x < 1$$



$$I_P = (0, 1)$$

$$I_M = (-\infty, \infty)$$

$$S = (0, 1)$$

5. ASINTOTI

$$D = (-\infty, +\infty)$$

$$\lim_{x \rightarrow -\infty} -x^3 + x^2 = -(-\infty)^3 + (-\infty)^2 = +\infty + \infty =$$

$$\lim_{x \rightarrow +\infty} -x^3 + x^2 = \lim_{x \rightarrow +\infty} x^3 (-1 + 1/x) = +\infty (-1)$$

$$m = \lim_{x \rightarrow \pm\infty} \frac{-x^3 + x^2}{x} = \lim_{x \rightarrow \pm\infty} \frac{x^2 (-1 + 1/x)}{x} = +\infty (-1)$$

6. DISSEGNO



